

Performance Testing

Date	15 March 2026
Team ID	XXXXXX
Project Name	Flood Risk Prediction using XGBoost
Maximum Marks	5 Marks

Step 1: Testing Overview

Testing Tool Used	Postman (API functional/performance checks)
Type of Testing	Load Testing
Target Module / API	/predict Flask API
Test Environment	Local Flask Server (Python)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Scenario	Users	Duration	Expected
1	Single prediction	1	60	Successful prediction
2	Concurrent predictions	10	60	Stable responses
3	Invalid input	5	30	Handled error
4	Continuous requests	20	120	No crash

Step 3: Results

Metric	Target	Actual	Status	Remarks
Avg Response	<2 s	0.4 s	Pass	Fast
Max Response	<5 s	1.3 s	Pass	Within limit
Throughput	20 req/s	24 req/s	Pass	Stable

Error Rate	<1%	0%	Pass	No errors
CPU	<80%	42%	Pass	Normal
Memory	<80%	38%	Pass	Normal

Step 4: Observations

Key Findings: The Flask API responded consistently under expected load. XGBoost predictions were generated quickly with low latency.

Bottlenecks: None observed during normal testing.

Optimizations: Feature engineering performed before prediction and model loaded once at startup.

Step 5: Screenshots / Evidence

Step 5: Screenshots

Flood Risk Prediction

Monsoon Intensity

Climate Change

Coastal Vulnerability

Infrastructure

Drainage Systems

Dams Quality

Urbanization

Deforestation

Encroachments

Population Score

Wetland Loss

Siltation

Agriculture

Disaster Preparedness

Planning

Political Factors

 Predict

Risk Level

Flood Risk Prediction

10

10

10

10

10

10

10

10

10

10

10

10

10

10

10

10

 Predict

 **HIGH FLOOD RISK**

Risk Level



Risk: 80%

⚠ Flood risk is HIGH. Heavy rainfall and weak drainage may lead to flooding. Stay alert and avoid low-lying areas.

Flood Risk Prediction

5

5

5

5

5

5

5

5

5

5

5

5

5

5

5

5

 Predict

 **SAFE AREA**

Risk Level



Risk: 20%

✓ Flood risk is LOW. Conditions are stable and no immediate flood threat is detected.

